

Smart Street Lights Using IoT – Motion Sensor Based Street Lighting System

1. Project Title

Smart Street Lights Using IoT and Motion Sensors for Energy-Efficient Street Lighting

2. Objective

To automatically control street lights based on vehicle/pedestrian movement using motion sensors, reducing energy consumption and enabling remote monitoring through IoT.

3. Problem Statement

Traditional street lights remain ON throughout the night, leading to unnecessary power consumption. A smart street lighting system can detect motion and adjust brightness accordingly, saving energy and improving efficiency.

Components Required

Component	Quantity
Arduino Uno	1
PIR Motion Sensor (HC-SR501)	1
LEDs (Street Lights)	4
220Ω Resistors	4
Breadboard	1
Jumper Wires	As required
USB Cable	1

Circuit Connections

PIR Sensor

PIR Pin Arduino Pin

VCC 5V

GND GND

PIR Pin Arduino Pin

OUT D2

LEDs

LED Arduino Pin

LED1 D8

LED2 D9

LED3 D10

LED4 D11

Connect a 220Ω resistor in series with each LED.